Project Proposal: quickImage

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EE/CS 147

Required Libraries and Frameworks

- **OpenCV**: for image processing and applying filters. Aid with background removal and HEIC support. Allowing for computer vision techniques for advanced filters and background removal.
- **CUDA Toolkit**: GPU acceleration and running the C++ kernels.
- **std_image / stb_image_write**: for loading JPGs and PNGs into raw byte arrays to copy to the GPU (device memory) and writing the output back to disk.
- **Argparse** (C++): for parsing command line arguments such as flags.

Potential Risks or Challenges

- **Background Removal**: Implementing accurate background removal can be challenging, especially for complex images. I may focus on portraits for this feature. I believe I can use convolution for this technique, but it may require experimentation and fine-tuning to achieve good results.
- **HEIC Support**: Most phones now save photos in HEIC format, so I want to support this, but JPG and PNG will be my priority, then I will add HEIC support if I have time.
- **Warp Divergence**: Using filters such as blur and edge detection will lead to warp divergence due to necessary conditional statements. Requiring optimization and experimentation to achieve good performance.
- **Boundaries and Edge Cases**: Handling blur where the pixel doesn't have a nearby pixel to sample from. Handling the padding in parallel threads will be tricky and may cause artifacts.

Plan/Outline for Project

Gist

quickImage is a CLI-based tool with GPU-accelerated image filters.

Motivation

I find myself opening Photoshop or Lightroom for very basic edits to my photos. I want a tool that can do those basic edits without the overhead of opening a large application and the acceleration of GPUs. quickImage will be a command line tool that can apply filters to images using the GPU for acceleration.

Inspired by the ease of use of ffmpeg, quickImage will have a simple syntax for applying filters to images. For example, to increase the brightness of an image, you could run:

```
quickImage input.jpg --brightness 0.2 --contrast 0.1
```

Features

Basic Features

- GPU acceleration for faster processing. Maybe have a live preview of the filter being applied to the image if you are using the CLI menu.
- Basic filters: brightness, contrast, saturation, hue, etc.
- Batch processing: apply filters to multiple images at once.
- Support for common image formats: JPEG, PNG, HEIC, etc.

Advanced Features

- Background detection and removal.
  - Remove background from portraits using edge detection and segmentation techniques.
  - Return a PNG with a transparent background.
  - Add a glow effect to the edges of the subject for a more artistic look.
- Filters.
  - Apply LUTs (lookup tables) for color grading and creative effects.
  - Advanced filters: blur, sharpen, gaussian blur, edge detection, etc.
  - Add Fuji Film's film simulation filters to give photos a classic look including film grain, color grading, and tone curve adjustments.
  - Add a filter that can mimic the look of vintage cameras, such as the Holga or Diana, which are known for their unique lens distortions and light leaks.
- Video support: apply filters to videos as well as images.
  - This would be a stretch goal, but it would be a great feature to have. It would allow users to apply filters to their videos in the same way they do with images.